

Anshul Patel

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EDUCATION

North Carolina State University, Raleigh, NC || *Master of Computer Science* Aug 2021 – Dec 2022
Courses: Software Engineering, Database Management System, Design and Analysis of Algorithms CGPA: 4.0
Nirma University, Gujarat, India || *Bachelor of Technology in Information Technology* Aug 2016 – May 2020
Courses: Machine Learning, Deep Learning, Probability Statistics and Numerical Analysis GPA: 8.57

SKILLS

Languages: TypeScript, JavaScript, Python, Java, PostgreSQL, NoSQL, SQL, C++
Cloud & Infrastructure: AWS CDK, Lambda, DynamoDB, S3, SQS, SNS, CloudFormation, KMS, CloudWatch, ECS Fargate Clusters, Docker, CI/CD Pipelines, Deno, Redis
Framework & Tools: Node.js, Express.js, Next.js, React, Supabase, Jest, Git, Github, REST APIs, GraphQL, Claude Code, Kiro, Gemini

WORK EXPERIENCE

Software Development Engineer II, Amazon Web Services, New York Oct 2024 – Apr 2026

- Architected [Kiro](#)'s complete metering and billing infrastructure from zero, including API Gateway ingestion, DynamoDB aggregation with stream-based triggers, EventBridge-scheduled limit evaluation, and batch delivery to AWS Metering, scaling to 20,000 developers at re:Invent 2025 with zero billing incidents on launch day.
- Engineered a high-frequency throttling service using Lambda and Redis Serverless to enforce token usage limits across 5-minute and 1-hour windows, achieving p50 latency under 1ms after benchmarking Redis over DynamoDB's transactional write latency at throttle-check scale.
- Owned onboarding flows and 6 third-party integrations end-to-end as responsible for the full lifecycle of AWS's first autonomous AI development agent.

Software Development Engineer I, Amazon Web Services, Seattle Feb 2023 – Sept 2024

- Architected materialized view processing enhancements including reverse relationships, one-to-many views, and V2 rearchitecture with resource association integration, enabling complex document aggregation and resource link traversal for [AWS CodeCatalyst](#) teams while improving performance by 40%.
- Established full ownership of business metrics infrastructure by implementing cross-account S3 replication, operational tooling, monitoring, and event processing integration, ensuring reliable data pipeline operations for CodeCatalyst's business intelligence capabilities.
- Led operational excellence and mentorship initiatives through weekly 1:1s with new engineers, ORR/COE documentation, and intern project support including end-to-end tracing dashboard development, achieving 62% operational backlog reduction.

Software Development Engineer Intern, Amazon Web Services, Seattle May 2022 – Aug 2022

- Utilized Typescript to develop a JSON Schema Linter CLI, which provided authors with guidance on best practices and resulted in a 70% reduction in review labor hours.
- Developed a Java SDK and integration tests for the AWS CodeCatalyst Eventing System.

Associate Engineer, Cloudoffis, Gujarat, India May 2020 – May 2021

- Transformed the classification model applying Active Learning techniques to detect and sustain both concept and data drift. Saved time by eliminating the need to retrain the model from scratch.
- Implemented dynamic multiprocessing in the pipeline, achieving a reduction in process time by 30%.

Machine Learning Intern, Knowarth Technologies, Gujarat, India Jan 2020 – May 2020

- Developed a multi-class classification model applying Ensemble techniques for over 1000 unstructured SMSF documents across 30 classes, attaining over 95% accuracy.

Project Intern, Knowarth Technologies, Gujarat, India May 2019 – Jun 2019

PROJECTS

(Audio Analysis) [Playback Attack Detection for Speaker Verification Systems](#)

- Analyzed the importance of acoustic speech features namely MFCC, IMFCC, RFCC, and LFCC extracted from ASVspoof 2017 dataset in classifying genuine and spoof audio signals to avoid Playback Attacks on Speaker Verification Systems.

(Full-Stack Web App) [Lekha - Portfolio Tracker](#)

- Built a community-driven portfolio showcase for the Indian investing community, supporting NSE, BSE, NASDAQ, NYSE, and Crypto with a social follower model for portfolio discovery, using Next.js 14, TypeScript, Supabase PostgreSQL, Tailwind CSS, and serverless Deno Edge Functions.
- Developed end-to-end using Claude as an AI development partner for code generation, schema design, and architectural decision-making, demonstrating production AI-assisted engineering across the full stack.